1

```
void reverse(Node *L){
    Node *p = L->next;
    Node *q;
    L->next = NULL;
    while(p){
        q = p->next;
        p->next = L->next;
        L->next = p;
        p = q;
    }
}
```

2

```
Node *middle(Node *head){
    Node *p = head;
    Node *q = head;
    for(int i = 1 ; p ; i++){
        p = p->next;
        if(!(i & 1)) q = q->next;
    }
    return q;
}
```

3

```
int length(Node *L){
    int n = 0;
    Node *p = L;
    while(p){
        p = p->next;
        n++;
    }
    return n;
}

Node *common(Node *L, Node *M){
    int l = length(L), m = length(M);
    Node *p = L;
    Node *q = M;
    while(l > m){
        p = p->next;
        l--;
    }
    while(m > l){
        q = q->next;
        m--;
    }
    while(p && q){
        if(p == q) return p;
```

```
        p = p->next;
        q = q->next;
    }
    return NULL;
}
```